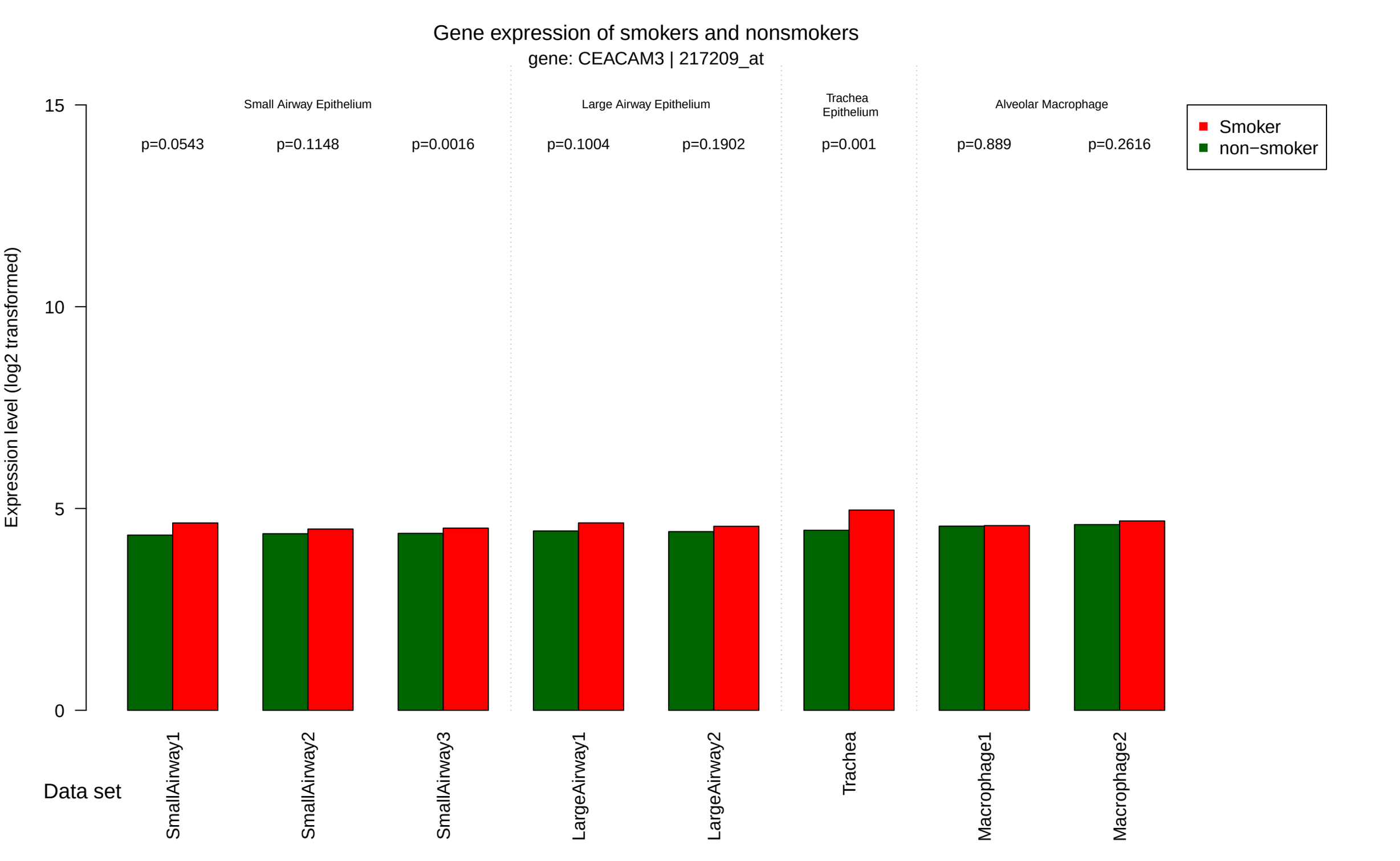
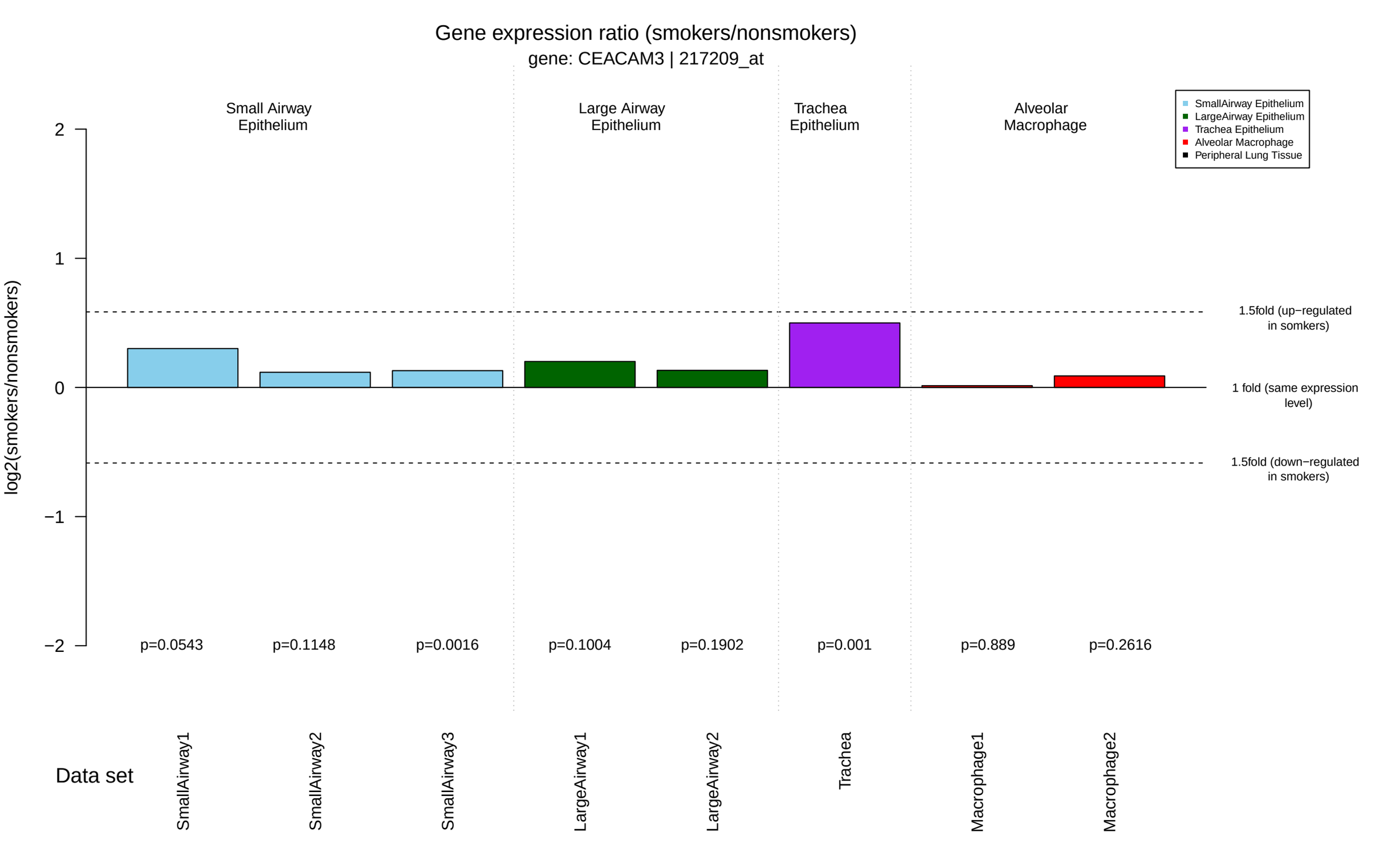
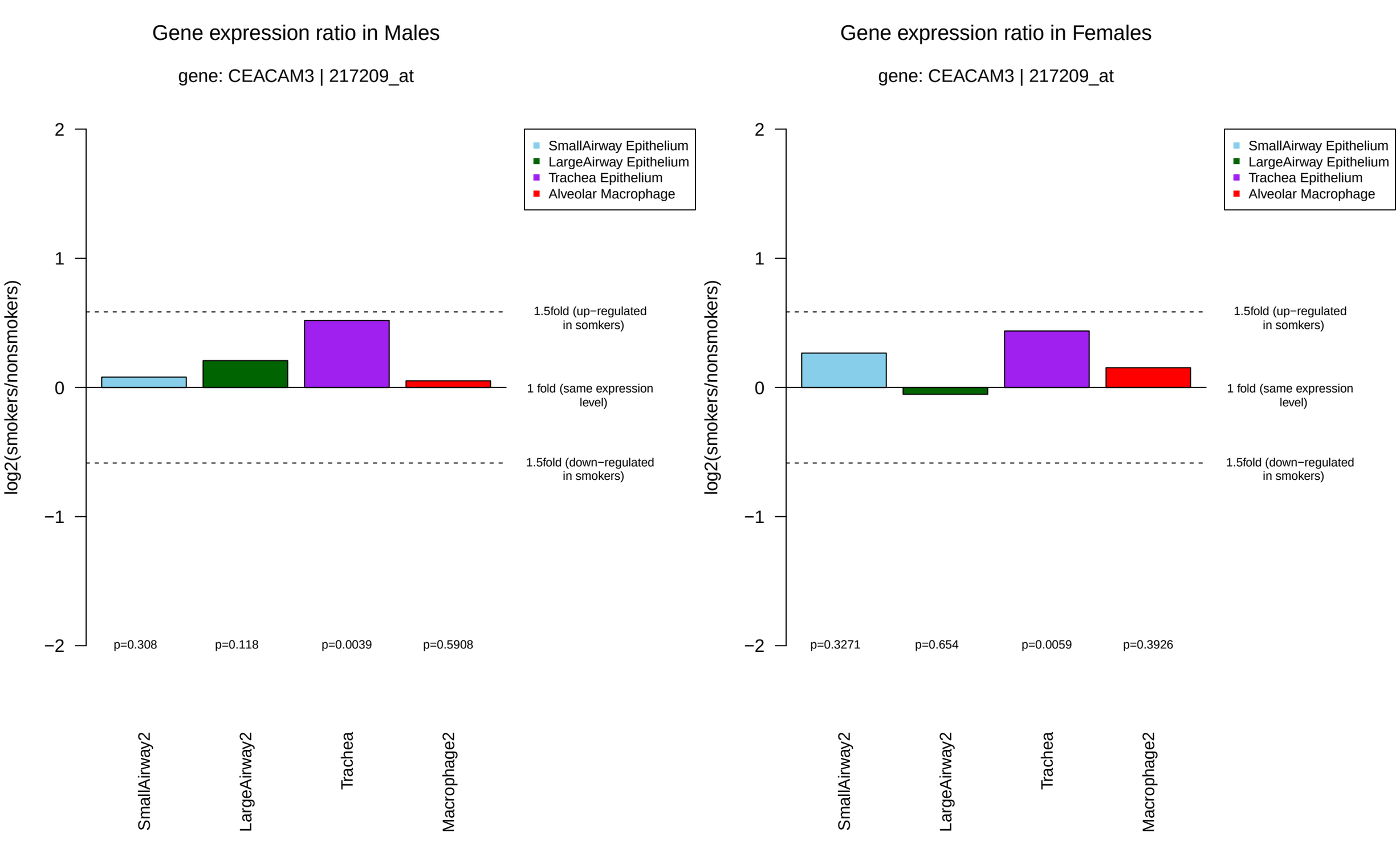
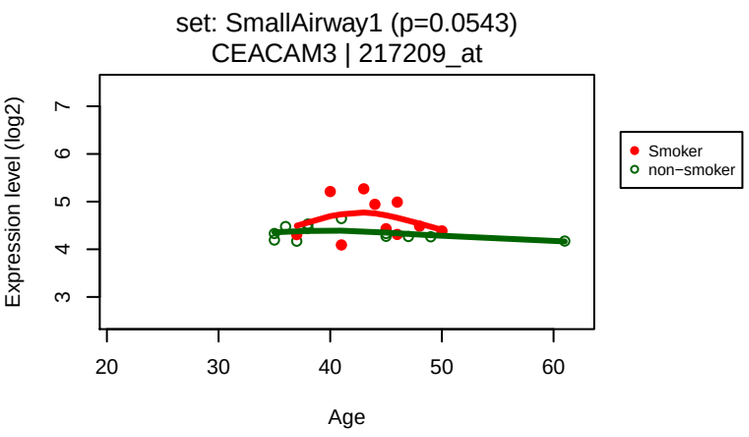
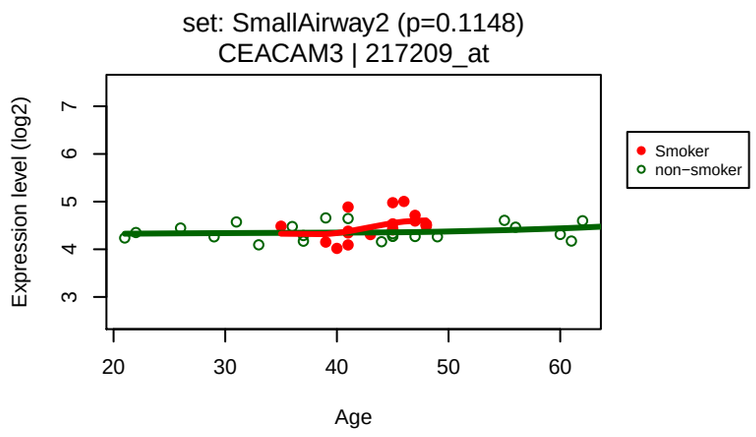
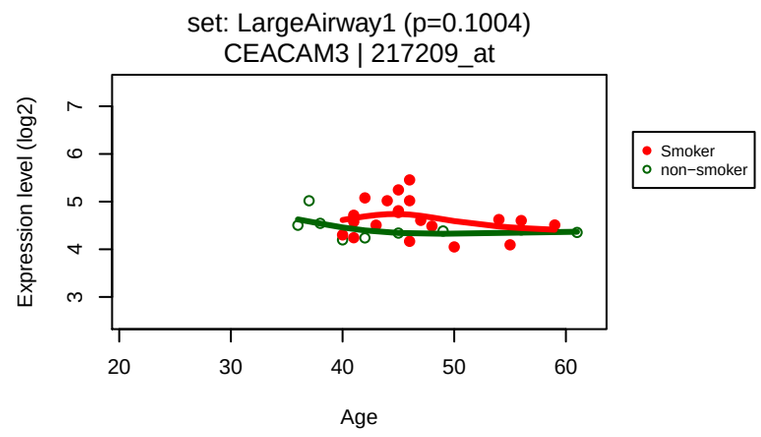

Gene expression of smokers and nonsmokers
gene: CEACAM3 | 217209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0543
p=0.1148
p=0.0016
p=0.1004
p=0.1902
p=0.001
p=0.889
p=0.2616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

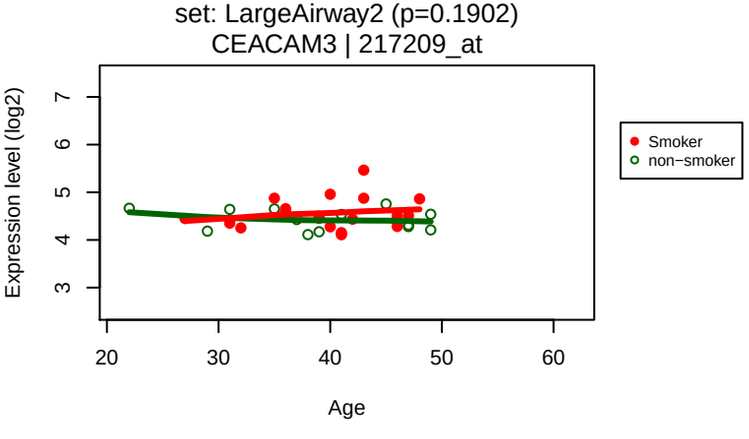

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM3 | 217209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0543
p=0.1148
p=0.0016
p=0.1004
p=0.1902
p=0.001
p=0.889
p=0.2616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

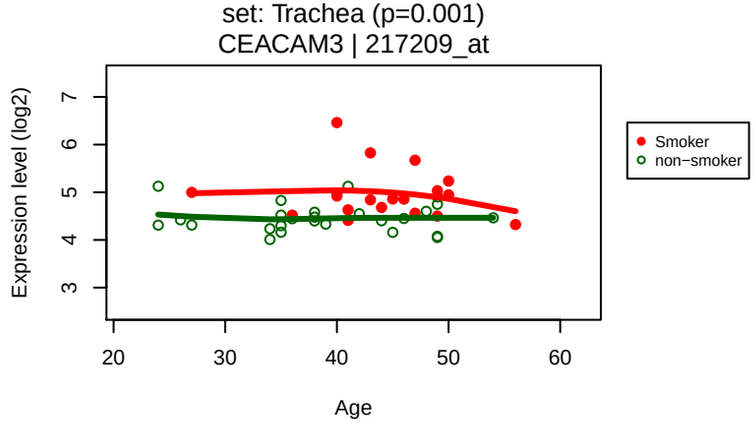

Gene expression ratio in Males
gene: CEACAM3 | 217209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.308
p=0.118
p=0.0039
p=0.5908
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM3 | 217209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3271
p=0.654
p=0.0059
p=0.3926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

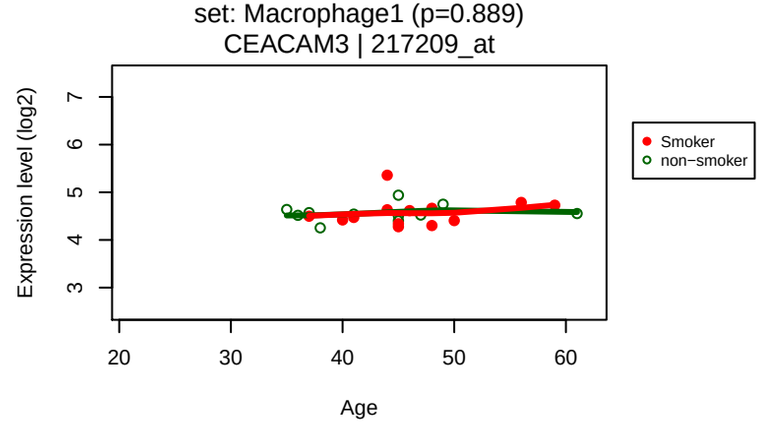

set: SmallAirway1 (p=0.0543)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker

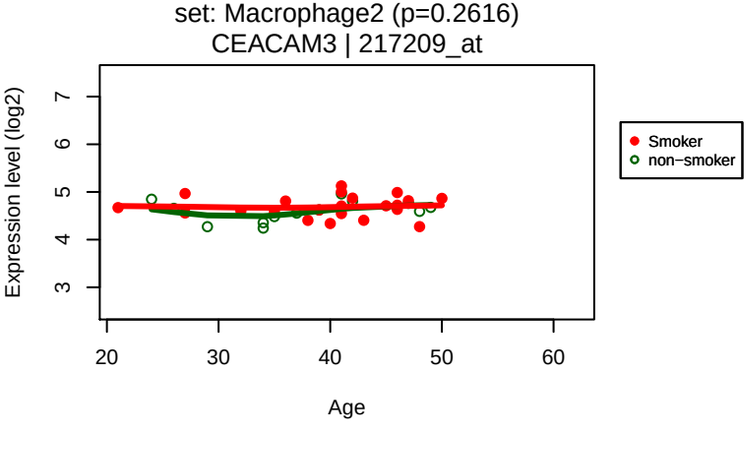

set: SmallAirway2 (p=0.1148)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker

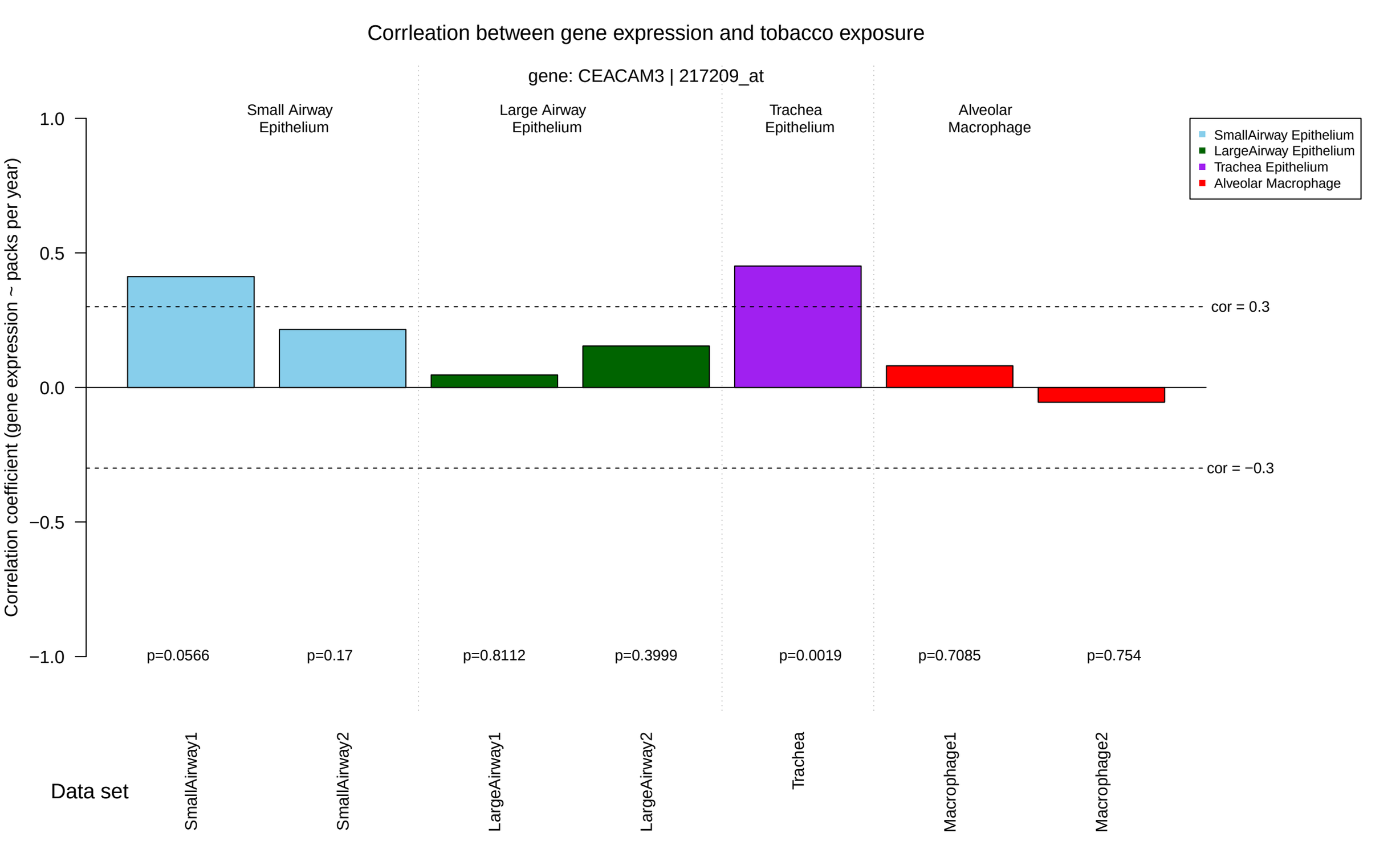

set: LargeAirway1 (p=0.1004)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1902)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.889)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2616)
CEACAM3 | 217209_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM3 | 217209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0566
p=0.17
p=0.8112
p=0.3999
p=0.0019
p=0.7085
p=0.754
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2